

What in the myths predicts cooperation? September beside frontier

Each cell: the lens's status (colour + symbol) and headline number. A cell spanning several rows is a pooled estimate over those rows; family lines inside it are the lens's per-family rows.

		1 Own first story = plan	2 Read story → next send	3 Read story → next story amount	4 Words copied from read story	5 Moral spreads within family	6 Moral spreads across families	7 Matching norms → cooperation	8 Drift toward consistency	9 Consistency → cooperation	10 Stories add beyond past moves	11 Morals follow play	
		R4, round-1 senders, myth→game k/n sent exactly the \$ named · \$ per	R3, 8-agent myth→game \$ sent per \$ in shown myth	8-agent myth→game \$ written per \$ in shown myth	adoption, shown – unseen percentage points	label uptake, myth→game shown – unseen, pts	label uptake, myth→game shown – unseen, pts	H1 primary: 5 norm measures × 3 outcomes = 15 tests	judge flag share, rd 1 → 8–10 (kw = keyword share change)	H3, agent FE, judge flag Δ send/5 · Δ return share	held-out R ² gain, all myth features (ridge); ≤ = upper CI	full send → P(next myth 'be generous'), Δ prob.	
SEPTEMBER · Sonnet 4.5 / GPT-5 Nano / Gemini 3.7 Flash · 156 myth runs													
All settings pooled (lens's headline strata)		✓ 44/62 sent the named \$ \$0.67/\$ [0.54, 0.79] · 37 runs Sonnet 32/40 · \$0.85 (yes) GPT 12/22 · \$0.48 (yes) Gemini 50/50 (ceiling)	○ -0.00 [-0.10, 0.10] 45 runs · placebo +0.06 Sonnet +0.00 (null) GPT -0.04 (null) Gemini: ceiling	✓ +0.07 [0.03, 0.12] 45 runs · placebo -0.03 Sonnet +0.09 (yes) GPT +0.09 (yes) Gemini: ceiling	not pooled across settings by the lens	not pooled across settings by the lens	not pooled across settings by the lens	✓ 2-ag: 1/15 Holm rule match → gap +0.019 8-ag: 0/15 Holm · 2 raw p	✓ Sonnet judge +.54, kw +.66 GPT judge +.56, kw +.34 Gemini judge +.24, kw +.18	○ send -.006 ±.034 return -.005 ±.009 156 runs, all families	○ send -0.000 (≤ 0.000) null return +0.000 (≤ 0.000) low-n Δ send -0.000 (≤ 0.000) null 111 runs, CV by run	✓ +0.16 [0.11, 0.21] all runs, run FE	
2-agent one model	Sonnet dyads	⊖ 3/4 sent the named \$ too few for \$ per \$ 4 runs · Sonnet+GPT Sonnet 3/4 (low-n) GPT 0 named (low-n) Gemini 5/5 (ceiling)	⊖ dyads: shown myth is always the partner's	⊖ dyads: shown myth is always the partner's	✓ +9.6 [7.6, 11.5] pts ↑ 30 runs · same family per family, m→g / g→m: Sonnet +7.7 / +8.8 GPT +17.2 / +13.5 Gemini +6.4 / +3.7	✓ +9.4 [4.1, 14.7] pts ↑ 15 runs · no placebo (dyads) Sonnet +10.8 (low-n) GPT +2.5 (low-n) Gemini +15.0 (sugg.)	⊖ one family only	○ 0/15 Holm 10 null · 5 low-n	~ judge +0.48 (.50→.98) kw +0.67 (yes) · 10 runs	○ send +.008 ±.020 ret -.011 ±.046 · 10 runs	pooled only ↑	○ +0.14 [-0.11, 0.38] agent-level 0/3 Holm, 0 raw	
	GPT dyads							○ 0/15 Holm 12 null · 3 low-n	~ judge +0.55 (.30→.85) kw +0.28 (yes) · 10 runs	○ send +.135 ±.106 ret -.014 ±.045 · 10 runs			
	Gemini dyads							▲ 0/15 Holm 3 null · 1 low-n · 11 ceiling	○ judge +0.25 (.75→1.00) kw +0.22 (null) · 10 runs	○ send ceiling ret -.014 ±.019 · 10 runs			
2-agent mixed pairs	Sonnet+GPT dyads	⊖ 2/4 sent the named \$ too few for \$ per \$ 4 runs · Sonnet+GPT Sonnet 1/2 (low-n) GPT 1/2 (low-n) Gemini 6/6 (ceiling)	⊖ dyads: shown myth is always the partner's	⊖ dyads: shown myth is always the partner's	✓ +2.1 [1.8, 2.5] pts ↑ 36 runs · cross-family myth→game +2.1 (yes) game→myth +2.2 (yes)	⊖ dyad partner is always the other family	✓ +10.4 [2.8, 17.3] pts ↑ 18 runs · no placebo (dyads)	○ 0/15 Holm 15 null	~ Sonnet: judge +.60 (sugg.) kw +.43 (yes), 24 runs GPT: judge +.67 (sugg.) kw +.40 (yes), 24 runs Gemini: judge +.19 (sugg.) kw +.19 (yes), 24 runs	○ Sonnet: send +.063 ±.069 ret +.046 ±.082 GPT: send -.027 ±.078 ret +.000 ±.043 Gemini: send ceiling ret +.016 ±.027	pooled only ↑	○ +0.35 [-0.03, 0.72] agent-level 0/6 Holm, 1 raw	
	Sonnet+Gemini dyads							○ 0/15 Holm · 2 raw p 8 null · 5 low-n					
	Gemini+GPT dyads							○ 0/15 Holm 10 null · 5 low-n					
8-agent one model	8 Sonnet	▲ 12/15 sent the named \$ \$0.80/\$ [0.59, 1.02] 8 runs · Sonnet+GPT Sonnet 11/12 · \$0.94 (ceiling) GPT 1/3 (low-n) Gemini 20/20 (ceiling)	⊖ +2.10 [-2.14, 6.33] 5 runs	⊖ -0.25 [-3.27, 2.76] 5 runs	✓ +4.9 [3.8, 6.0] pts 30 runs · same family per family, m→g / g→m: Sonnet +7.8 / +8.8 GPT +5.4 / +5.3 Gemini +1.2 / +1.0	~ +4.8 [2.6, 7.0] pts 15 runs placebo +3.9 (p .05): fails R3-adjusted +5.7 (yes)	⊖ one family only	○ 0/15 Holm 15 null	~ judge +0.55 (.42→.98) kw +0.85 (yes) · 10 runs	○ send -.026 ±.018 ret -.014 ±.033 · 10 runs	pooled only ↑	○ -0.00 [-0.09, 0.09] agent-level 0/3 Holm, 0 raw	
	8 GPT		⊖ -0.13 [-1.99, 1.72] 5 runs	⊖ -0.73 [-3.75, 2.28] 5 runs				○ 0/15 Holm 13 null · 2 low-n	✓ judge +0.57 (.30→.87) kw +0.17 (yes) · 10 runs	○ send +.043 ±.137 ret -.036 ±.073 · 10 runs			
	8 Gemini		▲ no estimate 5 runs	▲ no estimate 5 runs				▲ 0/15 Holm 4 null · 1 low-n · 10 ceiling	~ judge +0.29 (.71→1.00) kw +0.07 (null) · 10 runs	○ send ceiling ret -.010 ±.023 · 10 runs			
8-agent mixed	1 GPT + 7 Sonnet	✓ 27/39 sent the named \$ \$0.60/\$ [0.44, 0.76] 21 runs · Sonnet+GPT Sonnet 17/22 · \$0.82 (yes) GPT 10/17 · \$0.37 (ceiling) Gemini 19/19 (ceiling)	○ +0.00 [-0.11, 0.11] 30 runs · all families Sonnet -0.03 (null) GPT -0.04 (null) Gemini: ceiling	✓ +0.07 [0.02, 0.12] 30 runs · all families Sonnet +0.11 (sugg.) GPT +0.11 (yes) Gemini: ceiling	✓ +1.5 [1.3, 1.8] pts 60 runs · cross-family same family +6.3 (yes)	✓ +5.9 [3.1, 8.6] pts 30 runs placebo +2.3 (p .23): passes R3-adjusted +5.5 (yes)	○ +0.7 [-2.7, 3.7] pts 30 runs placebo -2.7 (p .23): passes R3-adjusted +0.3 (null)	○ 0/15 Holm 15 null	✓ Sonnet: judge +.51 (yes) kw +.77 (yes), 30 runs GPT: judge +.53 (yes) kw +.35 (yes), 60 runs Gemini: judge +.27 (yes) kw +.19 (yes), 30 runs	○ Sonnet: send +.001 ±.027 ret +.004 ±.014 GPT: send -.039 ±.068 ret -.008 ±.015 Gemini: send ceiling ret +.014 ±.017	pooled only ↑	✓ +0.21 [0.15, 0.26] agent-level 0/12 Holm, 3 raw	
	2 GPT + 6 Sonnet							✓ 1/15 Holm, OPPOSITE sign score match → send -0.039					
	4 GPT + 4 Sonnet							○ 0/15 Holm · 1 raw p 14 null					
	1 Gemini + 7 GPT							○ 0/15 Holm · 1 raw p 14 null					
	2 Gemini + 6 GPT							○ 0/15 Holm 15 null					
	4 Gemini + 4 GPT							○ 0/15 Holm 15 null					
FRONTIER (main set) · Opus 5 / GPT-5.6 Sol / Gemini 3.1 Pro · 106 myth runs													
All settings pooled (lens's headline strata)		✓ 50/52 sent the named \$ \$0.85/\$ [0.68, 1.02] · 27 runs Opus 19/21 · \$0.61 (sugg.) Sol 31/31 · \$1.00 (yes) GeminiPro 37/37 (ceiling)	▲ -0.01 [-0.04, 0.01] 20 runs · placebo -0.00 Opus: ceiling Sol -0.02 (ceiling) GeminiPro: ceiling	✓ +0.13 [0.05, 0.21] 20 runs · placebo +0.03 Opus +0.08 (null) Sol +0.13 (null) GeminiPro -0.01 (ceiling)	not pooled across settings by the lens	not pooled across settings by the lens	not pooled across settings by the lens	○ 2-ag: 0/15 Holm 8-ag: 0/15 Holm	✓ Opus judge +.24, kw -.01 Sol judge +.27, kw -.01 GeminiPro judge +.58, kw +.04	○ send -.003 ±.009 return -.005 ±.005 106 runs, all families	○ send +0.000 (≤ 0.000) null return +0.005 (≤ 0.011) sugg. Δ send -0.001 (≤ 0.004) null Opus return +0.039 (sugg.) 86 runs, CV by run	✓ +0.53 [0.29, 0.77] all runs, run FE DeepSeek +0.33 (sugg.)	
2-agent one model	Opus dyads	⊖ 4/4 sent the named \$ too few for \$ per \$ 4 runs · Sol+Opus Opus 0 named (low-n) Sol 4/4 (low-n) GeminiPro 5/5 (ceiling)	⊖ dyads: shown myth is always the partner's	⊖ dyads: shown myth is always the partner's	✓ +8.7 [7.1, 10.3] pts ↑ 30 runs · same family per family, m→g / g→m: Opus +9.9 / +11.3 Sol +4.5 / +5.5 GeminiPro +7.8 / +12.9	✓ +9.4 [3.0, 16.3] pts ↑ 15 runs · no placebo (dyads) DeepSeek +4.8 (null) Opus +3.3 (low-n) Sol +5.0 (low-n) GeminiPro +20.0 (low-n)	⊖ one family only	⊖ 0/15 Holm 4 null · 10 low-n · 1 ceiling	~ judge +0.30 (.60→.90) kw +0.00 (low-n) · 10 runs	○ send -.012 ±.019 ret +.002 ±.036 · 10 runs	pooled only ↑	○ +0.36 [-0.15, 0.87] agent-level 0/3 Holm, 0 raw	
	Sol dyads								⊖ 0/15 Holm 4 null · 11 low-n	○ judge +0.07 (.75→.82) kw +0.00 (null) · 10 runs			○ send +.091 ±.160 ret +.016 ±.049 · 10 runs
	GeminiPro dyads								⊖ 0/15 Holm 4 null · 10 low-n · 1 ceiling	✓ judge +0.55 (.35→.90) kw +0.10 (null) · 10 runs			✓ send ceiling · lens-wide .39 ret +.038 ±.024* · 10 runs
2-agent mixed pairs	Opus+Sol dyads	⊖ 8/8 sent the named \$ too few for \$ per \$ 8 runs · Sol+Opus Opus 3/3 (low-n) Sol 5/5 (low-n) GeminiPro 6/6 (ceiling)	⊖ dyads: shown myth is always the partner's	⊖ dyads: shown myth is always the partner's	✓ +3.9 [2.9, 5.2] pts ↑ 36 runs · cross-family myth→game +4.5 (yes) game→myth +3.4 (yes)	⊖ dyad partner is always the other family	○ +7.7 [-0.2, 15.9] pts ↑ 18 runs · no placebo (dyads) DeepSeek +5.1 (null)	⊖ 0/15 Holm · 1 raw p 4 null · 10 low-n	✓ Opus: judge +.15 (null) kw -.03 (low-n), 24 runs Sol: judge +.29 (sugg.) kw -.08 (null), 24 runs GeminiPro: judge +.58 (yes) kw +.03 (low-n), 24 runs	○ Opus: send -.007 ±.013 ret -.004 ±.022 Sol: send +.002 ±.029 ret +.003 ±.016 GeminiPro: send ceiling ret +.020 ±.039	pooled only ↑	○ +0.64 [-0.58, 1.86] agent-level 0/6 Holm, 0 raw	
	Opus+GeminiPro dyads												⊖ 0/15 Holm 5 null · 10 low-n
	GeminiPro+Sol dyads												⊖ 0/15 Holm 5 null · 10 low-n
8-agent one model	8 Opus	✓ 30/32 sent the named \$ \$0.81/\$ [0.58, 1.04] 10 runs · Sol+Opus Opus 13/15 · \$0.58 (ceiling) Sol 17/17 · \$1.00 (low-n) GeminiPro 20/20 (ceiling)	▲ no estimate 5 runs	⊖ +0.38 [-0.21, 0.96] 5 runs	✓ +11.2 [9.1, 13.5] pts 30 runs · same family per family, m→g / g→m: Opus +22.8 / +14.8 Sol +8.7 / +6.8 GeminiPro +7.4 / +6.5	✓ +6.8 [3.6, 10.2] pts 15 runs placebo +0.6 (p .53): passes R3-adjusted +6.5 (yes) DeepSeek +4.2 (sugg.)	⊖ one family only	○ 0/15 Holm · 1 raw p 9 null · 5 low-n	✓ judge +0.27 (.62→.89) kw +0.02 (low-n) · 10 runs	○ send -.000 ±.007 ret -.010 ±.011 · 10 runs	pooled only ↑	✓ +0.57 [0.25, 0.90] agent-level 0/3 Holm, 0 raw	
	8 Sol		▲ -0.02 [-0.09, 0.05] 5 runs	⊖ +0.08 [-0.01, 0.17] 5 runs				○ 0/15 Holm 14 null · 1 low-n	✓ judge +0.35 (.45→.80) kw +0.12 (yes) · 10 runs	○ send -.011 ±.021 ret -.003 ±.013 · 10 runs			
	8 GeminiPro		▲ no estimate 5 runs	▲ -0.03 [-0.12, 0.07] 5 runs				⊖ 0/15 Holm 5 null · 10 low-n	✓ judge +0.74 (.25→.99) kw -0.02 (low-n) · 10 runs	○ send ceiling ret -.016 ±.020 · 10 runs			
8-agent mixed	2 GeminiPro + 3 Opus + 3 Sol	⊖ 8/8 sent the named \$ too few for \$ per \$ 5 runs · Sol+Opus Opus 3/3 (low-n) Sol 5/5 (low-n) GeminiPro 6/6 (ceiling)	▲ no estimate 5 runs · all families Opus: ceiling Sol: ceiling GeminiPro: design	⊖ +0.13 [-0.15, 0.41] 5 runs · all families Opus +0.06 (low-n) Sol +0.22 (low-n) GeminiPro +0.09 (design)	~ same family +7.6 to +23.4 cross-family +4.3 to +10.4 5 runs · 16 pair*order rows, each Holm 1.0; no pooled row	⊖ +10.8 [-12.9, 28.7] pts 5 runs placebo +14.6 (≥ effect): fails GeminiPro: n/a (2 agents)	⊖ +13.1 [6.0, 20.0] pts 5 runs placebo +13.4 (≥ effect): fails R3-adjusted +13.6 (sugg.)	⊖ 0/15 Holm 5 null · 10 low-n	~ Opus: judge +.37 (sugg.) kw +.01 (low-n), 10 runs Sol: judge +.32 (sugg.) kw +.03 (low-n), 10 runs GeminiPro: judge +.47 (sugg.) kw +.07 (null), 10 runs	○ Opus: send -.010 ±.023 ret -.018 ±.015 Sol: send -.018 ±.040 ret -.009 ±.013 GeminiPro: send ceiling ret -.016 ±.031	pooled only ↑	~ +0.50 [0.03, 0.96] agent-level 0/3 Holm, 1 raw	
✓	yes (Holm p < 0.05)	~	suggestive (raw p < 0.05, not Holm)	○	no detectable effect	▲	can't test: ceiling (sends stuck at \$5)	⊖	too few runs / senders	⊖	not testable by design		

Statuses are the lenses' own, with Holm correction within each stratum (setting × family × task order); they can read stronger than the September README, which used lens-wide Holm (e.g. col 3 September +0.07 was not Holm-significant lens-wide). † dyads: partners share their game history and the unseen comparison myth comes from another run, so dyad word/moral matches are not transmission evidence (no future-myth placebo is possible). Cols 8–9 use the GLM judge flag (the only consistency measure live in both corpora); col 8 'yes' needs the DeepSeek judge to rise too, and frontier judges agree less per myth (κ 0.37 vs 0.54); the embedding score is not used as frontier drift evidence. Cells summarising several tests (cols 7, 9; mixed rows of 8; col 10): 'yes' if any lens row is yes, else the most common status. [a, b] = 95% CI (run-clustered); ± = CI half-width; * = Holm-significant. Moral labels and consistency flag: GLM-5.2 judge; DeepSeek as robustness (frontier DeepSeek served with hidden reasoning, not strictly comparable). Frontier = main set only (update set excluded). Game-only runs have no myths and are excluded. m→g = myth→game, g→m = game→myth task order; kw = keyword; snd = senders. Source rows for every cell: scorecard_cells.csv.